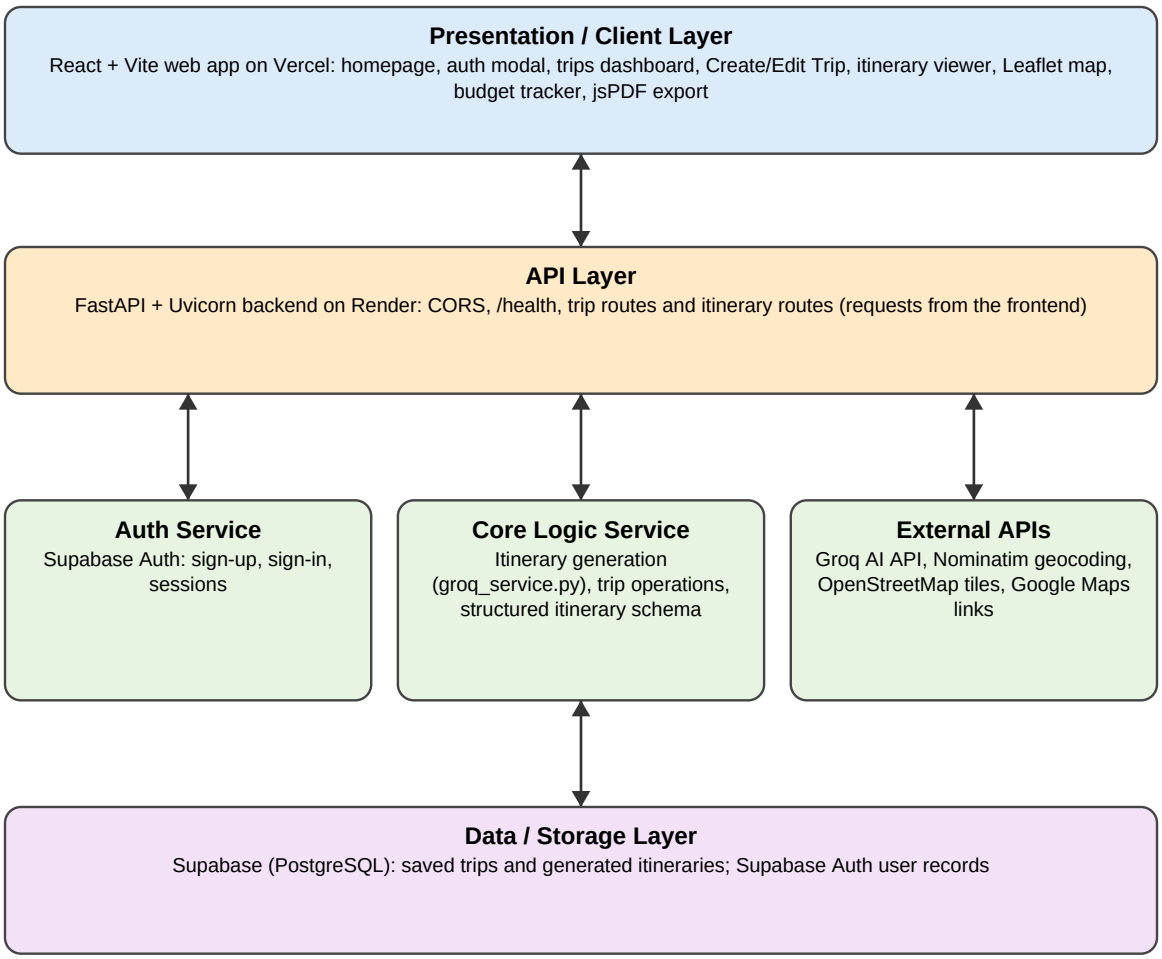
Solution Architecture

Date	23 October 2026
Team ID	[Your Team ID]
Project Name	TravelMate
Maximum Marks	5 Marks

Solution Architecture Diagram:

High-level view of the TravelMate architecture, mapped to the Presentation Layer, Application Layer, and Data Layer.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Shows the interface, collects trip details, displays the itinerary, map and budget, and exports the PDF.	React, Vite, JavaScript (JSX), CSS, Leaflet, jsPDF
API Layer	Receives frontend requests, applies CORS, and routes trip and itinerary operations.	Python, FastAPI, Uvicorn
Core Logic / External Services	Builds the AI request, parses the structured itinerary, geocodes place names and provides directions links.	Groq API, Nominatim, OpenStreetMap, Google Maps links
Database / Auth	Stores users and saved trips; authenticates users.	Supabase (PostgreSQL, Auth)